

# Murtaza S. Ahmed

Undergrad Engineer | Distributed Systems, ML Systems & Infra (Infrastructure focus)

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*Focusing on AI infrastructure, particularly GPU cluster orchestration, inference systems, model routing, resource allocation and workload scheduling.*

## Professional Experience

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| <b>Undergraduate Research Assistant (Edge AI Systems)</b><br>Department of Mechanical Engineering (MJCET)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>Dec 2025 -- Present</b><br><i>Hyderabad, TG</i>   |
| <ul style="list-style-type: none"><li>Selected for an institution-funded R&amp;D program under faculty supervision.</li><li>Built a complete edge AI system on NVIDIA Jetson Nano 4GB Developer Kit, integrating an IMX719 camera and soil sensors for real-time field operation.</li><li>Developed the on-device inference pipeline and executed real-time inference under compute and memory constraints.</li><li>Achieved 40–70 ms latency (14–20 FPS) using TensorRT (FP16) optimization.</li><li>Processed camera and sensor data continuously at 5–10 Hz with low-latency performance.</li><li>Designed a stable execution loop enabling reliable operation over 4–6 hour field deployments.</li></ul> |                                                      |
| <b>Undergraduate Researcher (R&amp;D Program)</b><br>Department of Mechanical Engineering (MJCET)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>March 2026 -- Present</b><br><i>Hyderabad, TG</i> |
| <ul style="list-style-type: none"><li>Design and implementation of an Adaptive Multi-Objective Policy Engine for Resilient Orchestration in Heterogeneous LLM Clusters.</li><li>Researching adaptive workload scheduling and hardware-aware resource allocation strategies for heterogeneous LLM inference systems.</li><li>Exploring fault-tolerant orchestration approaches for distributed AI workloads across mixed compute environments.</li></ul>                                                                                                                                                                                                                                                      |                                                      |
| <b>Builder &amp; Systems Architect</b><br>Verity (Decentralized Verification Platform)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>July 2025 -- Present</b><br><i>Hyderabad, TG</i>  |
| <ul style="list-style-type: none"><li>Designing and implementing a system to verify digital media provenance using blockchain and decentralized storage.</li><li>Integrating AI modules for manipulation detection and automated claim extraction.</li><li>Developing media provenance infrastructure using distributed storage systems including IPFS and Arweave.</li><li>Building concurrent ingestion pipelines and immutable audit logging mechanisms for ML-based verification workflows.</li></ul>                                                                                                                                                                                                    |                                                      |
| <b>Backend Infrastructure Engineer</b><br>Freelance Consulting                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Sept 2024 -- Present</b><br><i>Hyderabad, TG</i>  |
| <ul style="list-style-type: none"><li>Migrated client applications to containerized Linux deployment environments with automated CI/CD workflows.</li><li>Designed backend APIs and optimized PostgreSQL schemas for improved concurrency and deployment reliability.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                      |

## Projects

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| <b>Smart Agri Four-Legged Bot (SAFL-B)</b> <i>(C, Python, UART/Serial, I2C, GPIO, Hardware Control)</i>                                                                                                                                                                                                                                                          | <b>Dec 2025</b>  |
| <ul style="list-style-type: none"><li>Awarded 1st Place at the MAKEFORHYDERABAD 2-Day Makeathon for autonomous robotics systems integration.</li><li>Implemented deterministic UART/Serial communication protocols for multi-joint motor control.</li><li>Developed telemetry stabilization routines to reduce terrain-induced sensor and image noise.</li></ul> |                  |
| <b>Deterministic Workspace Provisioning</b> <i>(Bash, Linux, Arch Linux, Podman, Git)</i>                                                                                                                                                                                                                                                                        | <b>June 2025</b> |
| <ul style="list-style-type: none"><li>Built automated provisioning pipelines for lightweight Linux workstation deployment using Arch Linux and shell automation.</li><li>Configured container runtimes (Docker/Podman) and developer tools to deploy reproducible workspace layers.</li></ul>                                                                    |                  |

## Technical Skills

|            |                                               |
|------------|-----------------------------------------------|
| Languages: | C, Python, Bash, PostgreSQL, Git              |
| Systems:   | Linux, Arch Linux, Docker, Kubernetes, Podman |

## Education

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| <b>Bachelor of Engineering in Mechanical Engineering</b><br>Muffakham Jah College of Engineering & Technology (Osmania University)                        | <b>May 2028</b><br><i>Hyderabad, TG</i> |
| <ul style="list-style-type: none"><li>Relevant Coursework: Scientific Computing, Embedded Systems Programming, System Modeling &amp; Simulation</li></ul> |                                         |